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Review article

Renewable-storage sizing approaches for centralized and distributed renewable energy—A state-of-the-art review

Yuekuan Zhou

Sustainable Energy and Environment Thrust, Function Hub, The Hong Kong University of Science and Technology (Guangzhou), Nansha, Guangzhou 511400, Guangdong, China
Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong, China
HKUST Shenzhen-Hong Kong Collaborative Innovation Research Institute, Futian, Shenzhen, China
Division of Emerging Interdisciplinary Areas, The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong, China

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ABSTRACT

Low-carbon and sustainability transitions necessitate the intermediate bridge of battery for interconnections between renewables and demands. However, the empirical battery sizing approaches for both centralized and distributed energy systems lead to performance overestimation or underestimation, together with material and resource wastes. This study focuses on renewable-storage sizing approaches for centralized and distributed renewable energy systems to avoid battery capacity oversizing or under-sizing and resource waste. Renewable-storage sizing plays significant and dominant roles in techno-economic-environmental performances in long-term sustainability. Energy storages for both centralized and distributed energy systems are comprehensively reviewed, including both thermal and electrical energy systems. Roles of centralized and distributed energy systems are characterized in low-carbon transitions. In terms of renewable-storage sizing approaches, both centralized and distributed renewable-storage systems are characterized by 'U-value' approach and 'M-value' approach, respectively. Lastly, AI-assisted energy storage approach is also prospected with big data training surrogate model and sizing optimization. Research results indicate that distributed energy systems are more flexible in power sharing, transmission and distribution, together with fast load response, recovery and high energy resilience when suffering from natural disasters. Battery outpower stabilization and dynamic energy matching are principles for both centralized and distributed renewable-storage system designs. AI-assisted energy storage sizing approaches mainly include surrogate model development, performance prediction, and optimization. Research results can provide frontier guidelines on renewable-storage sizing approaches for both centralized and distributed energy systems, so as to promote the low-carbon and sustainability transitions.

1. Introduction

Global carbon neutrality transition imposes high requirement on renewable energy sources. Electrification and hydrogenation are main energy sources for carbon neutrality transition, while guidelines and economic incentives are required for implementation in practice [1]. Meanwhile, clean power transition can promote the Sustainable Development Goals [2], while carbon neutrality transformation requires economic investment [3]. Renewable energy integration will affect the power grid stability and resilience [4]. This necessitates battery storage for power stability when integrating renewables. As an intermediate bridge between renewables and energy consumptions, energy storages play significant roles in high renewable utilization, reliable power supply, grid power stabilization, and etc. However, with the accelerated deployment of renewable systems, associated battery systems will also increase due to the intermittence and instability of renewables. Energy storages play significant roles in wind and solar energy utilization, while cost reduction with energy and power dimensions is necessary [5]. Comparisons on different energy storages with different characteristics can assist the most appropriate technique selection for different scenarios [6]. Accurate battery sizing is essential to avoid oversizing or under-sizing, enabling the high-efficient resource utilization. However, considering the multi-criteria in renewable-battery integrated energy systems, there is no consistent approach to optimize the battery storage capacity.

Within traditional centralized energy systems, energy is generated

E-mail address: yuekuan.zhou@outlook.com.

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from large power plants, transmitted along the power grid for a long distance and then distributed to the consumers. However, the power supply from traditional centralized energy systems is inefficient, less flexible and less robust, with vulnerability to supply disruptions, the probability of grid instabilities and unexpected power outages. As intermediate components, energy storages connect both renewable energy sources and demand sides. The appropriate battery capacity determines both renewable sufficiency [7] and also dominates the load coverage [8]. Systematic review on sizing optimization of off-grid hybrid renewable energy systems can promote future research directions, like sizing optimization with energy management strategies [9]. Sizing methods and optimization techniques have been comprehensively reviewed on PV-wind-based hybrid renewable energy systems [10]. The comparison between different approaches indicates that, depending on different energy systems (either centralized or distributed energy systems), sizing optimization classifications are important for power supply reliability [10]. However, there is no systematic study focusing on optimal battery sizing for both centralized and distributed energy systems, leading to performance overestimation or underestimation, together with material and resource wastes.

Considering the significance of batteries, battery optimizations have been conducted, in terms of battery capacity and smart charging/discharging strategies. Optimal battery approaches mainly include directed search-based methods, probabilistic methods, and control strategies [11]. Astaneh et al. [12] optimized both sizing and control strategies of lithium-ion batteries within an off-grid renewable energy system. The optimization results can improve the economic performance and battery lifetime to 14.6 % and 78.4 %, respectively. Abdelshafy et al. [13] optimized the energy management strategy of a renewable-battery system. They indicated that the integration of pumped storage could effectively reduce the cost of energy by 22 %. However, both technical readiness (like smart charging and discharging) and market acceptance for battery storages have not been well investigated.

Along with the fast development of artificial intelligence and big data technologies, digital energy technologies have attracted researchers' interests. Based on the super learning capacity in nonlinear performance through support vector regression and backpropagation, machine learning plays significant roles in energy system design [14,18], control [15], efficient management [16], operations [17] and battery aging prediction [19,20]. In terms of power flow control, Xu et al. [15] studied a deep reinforcement learning approach to control the power flow into/from battery. The case study validates the effectiveness of deep reinforcement learning than model predictive control (MPC) in prediction efficiency and robustness. Regarding the battery energy management, Nyong-Bassey et al. [16] studied the adaptive PoPA-based energy management strategy based on machine learning. Results indicate that reinforcement learning-based adaptive power pinch shows the best performance. Furthermore, Zhou [20] developed a regression learner-based approach to dynamically predict the battery aging under different charging from renewables and different discharging for demand. Research results indicate the reliability and robustness of the proposed approach. However, roles of artificial intelligence (AI) in battery sustainability have not been systematically provided.

Within the battery-integrated energy systems, multiple techno-economic-environmental performances are investigated. However, these criteria conflict with each other, such as the minimum costs but with maximum carbon emissions [21]. Sunny et al. [22] studied an integrated system with optimal photovoltaic, wind, diesel generator and battery, in term of fuel costs, CO_2 emissions, and net present cost. The optimization results based on genetic algorithm (GA) and grey wolf optimizer (GWO) show the same results in system configuration. Das et al. [23] conducted the multi-objective optimisation of an electric vehicle (EV) integrated system. Results indicate that additional compensation to end electricity user and the electric vehicle owner is required to achieve the preset improvement in grid utilization. However, considering the high costs of battery, the tradeoff between battery aging and operational cost savings needs to be achieved. Zhou [24] studied both battery degradation costs and operating cost savings of an EV-integrated novel spatiotemporal energy network. Results indicate that internal trading cost can overwhelm the battery degradation costs, enabling the economic feasibility of the proposed spatiotemporal energy network. In order to further enhance the multi-stakeholders' profits, Zhou [25] studied dynamic energy pricing strategies for internal power trading between buildings, vehicles and power grid. The proposed pricing strategies can overcome the battery degradation cost and economically incentivize participation willingness of multi-stakeholders.

Based on the above literature review, scientific gaps can be summarized below:

1) there is no systematic study focusing on optimal battery sizing for both centralized and distributed energy systems. The parametrical analysis in the academia on battery capacities will lead to performance overestimation or underestimation, together with material and resource wastes.
2) as an intermediate energy bridge, battery storage plays significant roles in renewable self-sufficiency through charging and load service provision through discharging. In order to maximize the techno-economic performance of the battery, integrated renewable-storage-grid energy systems require the trade-off between operating cost savings and battery depreciation costs. However, both technical readiness (like smart charging and discharging) and market acceptance for battery storages have not been well investigated;
3) techno-economic performance analysis (TEA) is highly dependent on battery performance for renewable-storage-grid energy systems. Energy digitalization necessitates the implementation of machine learning in renewable and sustainable energy field. However, roles of artificial intelligence (AI) in battery sustainability have not been well studied.

This study is to address the above gaps. Research contributions and originalities are summarized below:

1) In order to guide the design on renewable-battery system, optimal battery capacity sizing approaches have been provided for both centralized and distributed energy systems. Both advantages and disadvantages of already existing optimal battery sizing approaches have been deeply analysed. Furthermore, innovative battery sizing approaches are proposed for both centralized and distributed energy systems.
2) both technical readiness (like technique maturity and smart charging/discharging) and market acceptance are provided on battery storages in both static and mobile forms, to help investors make decisions. Transition tendency towards distribution and electrification has been studied under the renewable-storage-grid framework.
3) roles of AI in battery storages have been analysed, in terms of surrogate model development, performance prediction and sizing optimization. Techno-economic Analysis (TEA) has been conducted for renewable-storage-grid energy systems with optimal sizing.

This study is organized as Section 2 for Research methodology. Section 3 shows the systematic literature review of renewable-storage-grid energy systems. Afterwards, renewable-storage sizing approaches are introduced in Section 4. Techno-economic Analysis (TEA) was conducted in Section 5 for renewable-storage-grid energy systems with optimal sizing. Outlook and recommendations are shown in Section 6. Lastly, conclusions are drawn in Section 7.

2. Methodology

A holistic literature review of renewable-storage sizing for centralized/distributed energy systems was conducted following the roadmap, as listed in Fig. 1. Renewable-storage-grid energy systems and renewable-storage sizing approaches are reviewed. Energy storages for centralized and distributed energy systems are comprehensively reviewed, including both thermal and electrical energy systems. Roles of centralized/distributed energy systems are characterized in low-carbon transitions. In terms of renewable-storage sizing approaches, both centralized and distributed renewable-storage systems are characterized by 'U-value' approach and 'M-value' approach, respectively. AI-assisted energy storage approach is also prospected with big data training surrogate model and sizing optimization with AI. Afterwards, techno-economic analysis (TEA) for renewable-storage-grid energy systems is conducted with optimal sizing. Outlook and recommendations are finally provided to pave path for upcoming studies.

3. Systematic literature review of renewable-storage-grid energy systems

In the academia, there are a lot of studies focusing on techno-economic-environmental performance analysis of integrative renewable-storage-grid energy systems. Table 1 summarized both centralized and distributed energy storages for high-efficient energy utilizations. Renewable-storage-grid energy systems mainly include centralized and distributed energy systems. Generally, the systems are in different forms, and research objectives can be mainly classified into technical (like renewable penetration and curtailment levels, grid loading, reliability of power supply, etc.), economic (line operating cost, levelized cost of energy, annualized cost) and environmental aspects (equivalent CO_2 emission). Research methods mainly include comparative analysis, Monte Carlo simulation, and optimization.

3.1. *Energy storages for centralized energy systems*

3.1.1. *Centralized thermal energy systems*

Centralized thermal storages are mainly designed for district heating [36], waste heat recovery from industry or data center [37], centralized solar thermal systems [38], and etc. Jimenez-Navarro et al. [39] studied roles of centralized and district heating systems. Results highlight the significance in thermal storages for both energy flexibility and efficiency improvement. The 4th generation district heating with thermal energy storages can improve the overall energy efficiency and minimize operating costs. Waste heat from data center can reduce both payback time and carbon emissions [40]. Thermal energy storage with various renewable integrations can reduce bypass loss and improve the energy use efficiency [41]. Furthermore, stakeholders' participation plays critical roles in energy flexibility for district heating. Initial economic investment, normal operation, investment and system acceptance are highly dependent on stakeholders. Renewable energy use in district heating grids is essential for incentivizing stakeholders' participation [42].

3.1.2. *Centralized electrical energy systems*

In addition to centralized thermal energy systems, centralized

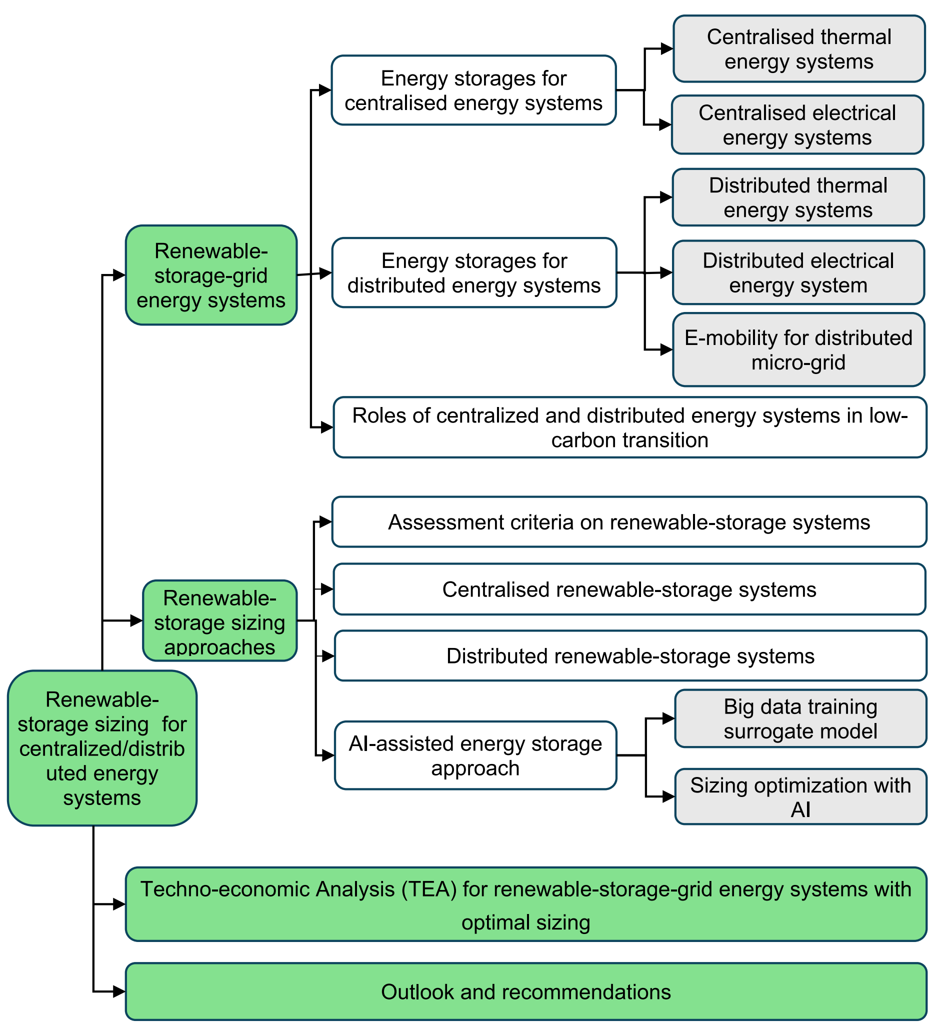

Fig. 1. Roadmap of renewable-storage sizing for centralized/distributed energy systems.

Table 1
Centralized and distributed energy storages.

System types	Studies	Systems	Objectives	Methods	Results
Centralized energy storage	Headley et al. [26]	Grid-battery storage	Renewable penetration and curtailment levels	Renewable curtailment on battery storage capacity	Renewable penetration ratio target of 60 % in 2030
	Kebede et al. [27]	Renewable-grid-battery	Suitable energy storage selection	Techno-economic and environmental impact analysis	Lithium-ion battery and thermal energy storage are suitable for seasonal energy storages.
	Sepulveda et al. [28]	Renewable-energy storage system	Energy capacity costs with storage durations	Case study	Energy capacity costs lower than US$20 kW/h can reduce electricity costs by over 10 %.
	Dowling et al. [29]	Renewables with Long-Duration Energy Storage	Operating cost	Case study	Batteries are for intra-day storage and long-duration storage can be used for season and multi-year storage.
	Hunter et al. [30]	Hydrogen, compressed air and pumped thermal storage	Levelized cost of energy and renewable energy penetration in power grid	Comparative analysis	The proposed fuel cells can decrease the levelized cost of energy by 13 %–20 % with over 80 % renewable penetration in power grid.
Distributed energy storage	Weckesser et al. [31]	Distributed PV-battery system	Grid loading and economic benefit	Optimization model	The proposed sizing and energy management strategy can reduce the grid loading by up to 58 % and ensure the economic benefit.
	Luo et al. [32]	Standalone energy system in Nan'ao Island	Reliability of power supply and annualized cost	Genetic algorithm and sequential simulation	The proposed sizing method and control strategy are cost-effective in reliable power supply with stable grid power.
	Abbassi et al. [33]	PV/Wind-battery system	Battery life cycle and power fluctuations	Monte Carlo Simulation.	Cumulative probability levels are used to determine the battery capacity.
	Bartolini et al. [34]	Renewable-driven community	Operating cost	Power-to-gas and power storage	Synergies on multi-energy system is the most cost-effective strategy.
	Zhou et al. [35]	Hybrid H_2-battery energy storage in building prosumers	Techno-economic performance	Parametrical analysis and energy management strategy	Synergies on battery and hydrogen systems can effectively reduce the operating costs.

electrical energy systems have also attracted widespread interests from both researchers and engineers. Li et al. [43] comprehensively reviewed battery pathway for carbon neutrality transition, including centralized stations, distributed electric vehicles (EVs), and large-scale hydrogen energy storages. They concluded that energy storage was essential for renewable markets. Grid service provision with battery to replace combined cycle gas turbines can effectively reduce greenhouse gas emissions [44]. Low-carbon grid with renewable integrations can reduce EV emissions, while advanced energy management strategies (EMS) and optimal battery storage are necessary for high-power EV charging [45]. Research results indicated that the levelized emissions of energy supply could be reduced with grid transformation and smart EMS.

In terms of renewable-EV charging systems with multi-battery storages, energy management strategies can effectively improve energy self-sufficiency and reduce the number of battery cycles [46,99]. PV-battery with energy sharing for community service provision can enable flexibility, reliability and climate resilience [47]. Agnew et al. [48] studied residential solar and storage for grid power supply. Smart grid architecture with district energy storages can provide sufficient energy flexibility to the local power grid [49].

3.2. Energy storages for distributed energy systems

Distributed energy systems can provide fast demand response, self-consumption and sustainability [50]. This section is to systematically review distributed thermal/electrical energy systems, together with E-mobility for distributed micro-grids.

3.2.1. Distributed thermal energy systems

Distributed electric vehicles, heat pumps and thermal energy storage with model predictive control can improve energy flexibility in according to hourly electricity pricing and climate change [51]. Seasonal energy storage for energy management in distributed energy systems can provide energy flexibility and climate adaptiveness [52]. Du et al. [53] reviewed phase change material (PCM) storages for industrial waste heat recovery and for distributed heat supply. The review indicates the high price of the thermal storage unit with the long transport distance, and the charging/discharging process is fast in the direct-contact system. Renewable energy systems with PCM thermal energy storage can ensure thermal energy supply stability in buildings with much less costs than the power storages (like battery and hydrogen) [54]. Zhou [55] studied demand response flexibility from passive PCM walls, BIPVs, and active air-conditioning systems. Results indicate the thermal regulation capacity of building envelopes for peak shaving and cost savings.

3.2.2. Distributed electrical energy systems

In addition to distributed thermal energy systems, distributed electrical energy systems have also been widely studied. Zheng et al. [56] studied distributed battery storages for energy sharing. They indicated that transactive energy and grid feed-in tariff dominated the energy sharing strategies. Kang et al. [57] comprehensively reviewed step-by-step studies on battery-based smart cities. They indicated that battery development mainly included sizing strategy, scheduling strategy and allocating strategy, as demonstrated in Fig. 2. Storage options and sizing of integrated renewable energy systems have been systematically reviewed for stand-alone applications [58]. A shared solar and battery storage system can enable the self-sufficiency of renewables in buildings, together with the decrease in energy consumption and the enhancement in grid independence [59]. Considering the replacement of coal power plants with flexibility, customer-sited energy storage can shift power for cost saving based on valley-to-peak price and reduce carbon abatement cost [60].

Furthermore, as a mobile energy storage, electric vehicles play essential roles in spatiotemporal energy sharing, high renewable penetration and grid power stabilization. Mobile EVs can support local power grids and the coordinated allocation of distributed generation resources can help reduce the operating costs [61]. EVs can achieve peak shaving [62], demand response, energy arbitrage, frequency regulation [63]. Solar power, batteries and green hydrogen on the off-grid mode can ensure power supply reliability with low carbon emissions [64]. Distributed EVs can provide supports for renewable-powered grid. The case study indicates that Vehicle-to-Grid (V2G) charging can supply 205 GW capacity and 14.7 ¢/kWh [65]. Yu et al. [66] comprehensively reviewed vehicle-to-grid interactions, in terms of power architectures and grid connection standards. Energy sharing and management based on EVs can help improve self-sufficiency and participants' cost savings [67]. Energy prosumers' coordination on distributed energy systems

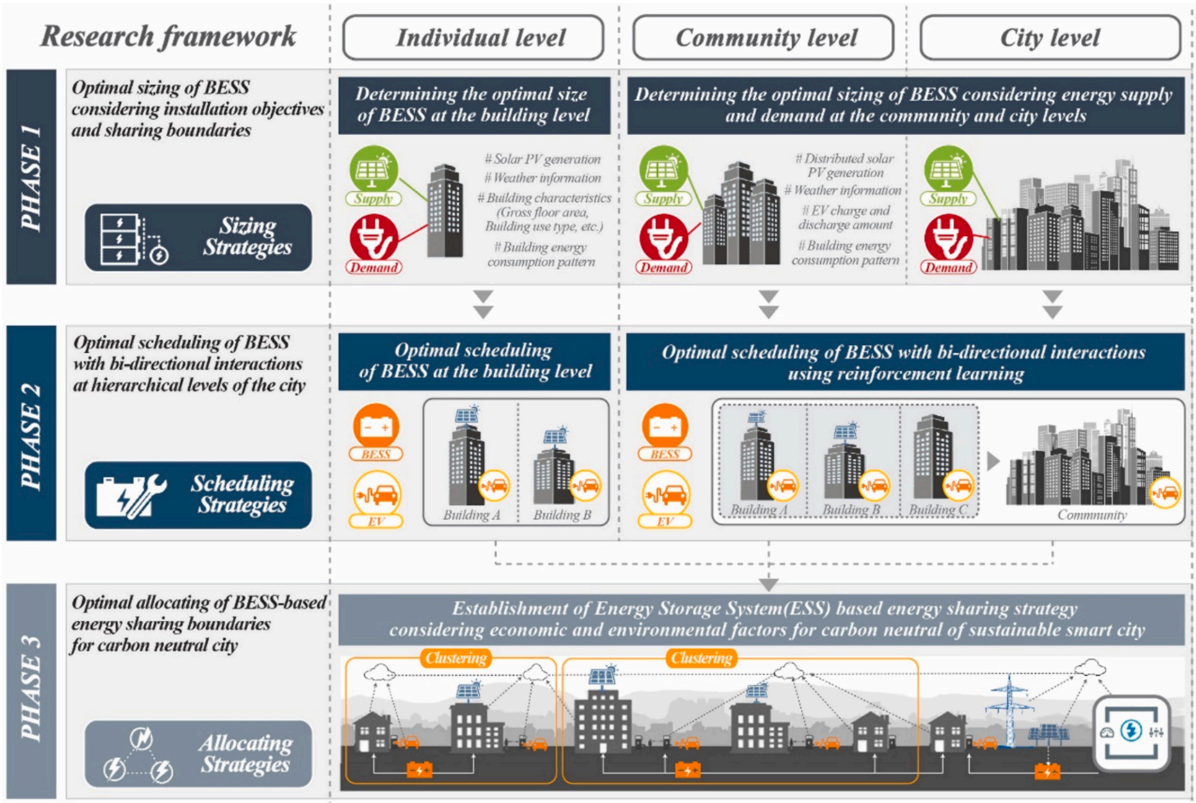

Fig. 2. Development of battery energy systems in smart cities [57]. Figure 2 is partially reprinted and partially redrawn from Kang et al. [57] with permission from Elsevier.

with a two-stage distributed algorithm can ensure the rational profit allocations [68]. Goop et al. [69] studied the PV-battery system in single-family houses with model development and profitability analysis. They indicated that market feedback needs to be considered in the modelling process. PV-EV-battery synergy can promote the distributed renewables in power systems, while users' acceptance in investment is necessary to be studied [70]. Furthermore, the second-life electric vehicle lithium-ion batteries can also support the distributed power grid [71] and distributed renewables in buildings [72]. However, insufficient subsidies, lack of funding and supporting policies are main challenges for decentralized renewable energy development [73].

3.2.3. E-mobility for distributed micro-grid

E-mobility can provide energy flexibility to communities and power grid [74], frequency stability [75], load reshaping, bidirectional Vehicle-to-Grid (V2G) service provision and load flexibility provision [76]. As shown in Fig. 3, on the one hand, from the building energy systems' side, the intermediate EVs can provide renewable penetration, demand coverage, grid independence with cost savings. On the other hand, from the power grid's side, the intermediate EVs can provide frequency stability, load reshaping, bidirectional V2G service provision

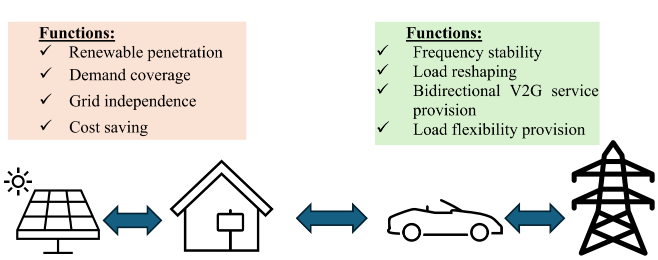

Fig. 3. Diagram on electric vehicles on micro-grid.

and load flexibility. The systematic review on e-mobility and buildings with distributed renewables paves pathways for sustainability [77]. Bayram et al. [78] studied storage capacity with Matrix-geometric algorithm. The vehicle-to-vehicle power transfer can support the Micro Grid charging station [79]. Jenn and Highleyman [80] analysed the upgrade of power grid on EVs. They concluded that 20 % circuits need upgrades and electric vehicles are important for local distribution networks. The deep EV deployment in market requires charging infrastructures to reduce the grid impact [81].

3.3. Roles of centralized and distributed energy systems in low-carbon transition

Centralized and distributed energy storages have attracted widespread researcher's attention [82]. Centralized storages can achieve more energy savings and distributed home batteries can enable energy flexibility for grid. Emrani-Rahaghi et al. [83] studied electric vehicles integrated residential energy hubs considering uncertainties on renewables and energy prices. The case study shows the operating cost saving potentials from EVs and thermal energy storages. For low-carbon transitions, optimal energy unit planning and energy efficiency improvement are critical. Firstly, optimal energy units planning can help reduce unnecessary energy storages or conversion processes [84,85]. Furthermore, the optimal energy unit locations can reduce power transmission losses [86]. Energy efficiency improvement in centralized and distributed energy systems mainly works throughout the entire energy supply chain, i.e., power supply [87], transmission, storage [90], distribution, and end-users [88].

Compared to centralized energy systems, distributed energy systems are more flexible in power sharing, transmission and distribution [91]. Furthermore, distributed energy systems can enable self-consumptions to reduce the energy storage capacity [92] and enable fast demand response [50] and recovery with high energy resilience when suffering from nature disasters [93–95]. By contrast, centralized energy systems show a higher energy efficiency, power supply reliability [96], and etc.

4. Renewable-storage sizing approaches

Renewable-connected, grid-connected, and demand-side battery deployment strategies are widely applied for sustainability transformations, while design principle, operation control and techno-economic analysis are quite different. Unlike centralized battery storages only considering power supply characteristic with uniformity factors for capacity sizing [97], the demand-side battery needs to simultaneously consider both power supply and energy demand characteristics. The provincial comparison in China indicates that, renewable-connected battery strategies are suitable in areas with abundant renewable resources, while the demand-side battery strategies are suitable in areas with limited renewables [98]. Solar photovoltaic-battery systems are comprehensively reviewed in optimal planning, challenges and new perspectives [99].

In this section, assessment criteria on renewable-storage technologies are provided (Section 4.1). Based on the criteria, renewable-storage sizing approaches are studied for both centralized and distributed systems (Sections 4.2 and 4.3). Lastly, AI-assisted energy storage approach is provided, including big data training surrogate model and battery capacity sizing optimization (Section 4.4).

4.1. Assessment criteria on renewable-storage technologies

Techno-economic and life cycle assessments on energy storage technologies are critical for capacity sizing. Fig. 4 demonstrates the power supply chain with production, transmission, distribution and end-users. Table 2 summarizes the optimization and assessment criteria of renewable-storage systems. The multi-criteria mainly include renewable penetration, battery capacity degradation and service life, levelised costs of electricity and heat. Rahman et al. [110] reviewed round-trip efficiency, cycle life, cycle length and levelized cost of electricity of various storage technologies, together with carbon footprint. Rawa et al. [111] optimized the battery storage capacity based on the objective to minimize the total operating cost of the grid.

Unlike centralized PV-battery-consumer systems that mainly focus on intermittent renewable energy, batteries in distributed prosumer-battery systems have to dynamically balance on-site renewable energy supply and energy demand [119], imposing challenges on battery capacity optimization. Furthermore, in terms of electrified lifecycle sustainable transformation, it is still questionable that whether a centralized or distributed energy system is the most effective solution or not.

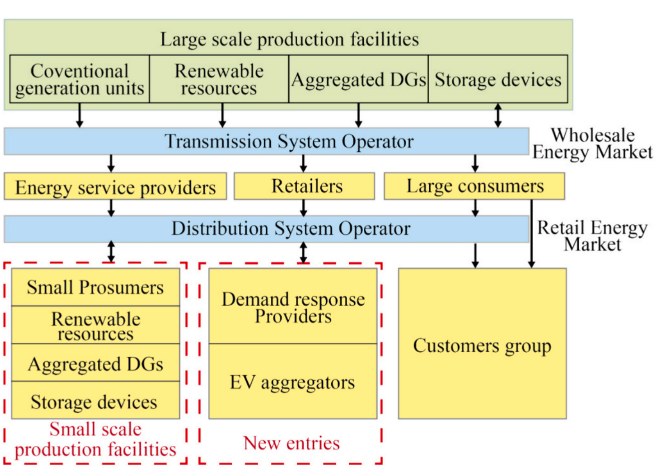

Fig. 4. Power supply chain with production, transmission, distribution and end-users.

Fig. 5 demonstrates the battery capacity sizing methods for both centralized PV-battery-consumer and distributed PV-battery-building systems. Regarding the centralized PV-battery-consumer system, the general principle is to stabilize the outpower from the battery with the battery capacity-dependent maximum power (P'_{max}). In terms of the distributed PV-battery-building system, the battery design needs to simultaneously consider both renewable supply and energy demand. The principle is to reduce the mismatched energy (i.e., E_{surp} and E_{short}) by designing the battery storage. Depending on the battery capacity, different magnitudes in mismatched energy can be achieved.

4.2. Centralized renewable-storage systems

Battery capacity of a centralized renewable energy system is optimized using the U-value method [97]. Table 3 summarizes the capacity sizing on centralized electrical energy systems. Generally, capacity sizing approaches mainly include parametrical analysis, single-objective and multi-objective optimizations. Al-Masri et al. [121] applied flow direction optimization algorithm to identify the optimal system configurations as hybrid PV/wind/pumped hydro storage (PHS) systems. Based on the existing energy systems, Sadeghibakhtiar et al. [122] conducted single-objective and multi-objective optimisations to ensure the energy supply reliability and reduce system costs.

4.3. Distributed renewable-storage systems

The new M-value approach has been proposed to size the battery capacity of distributed energy systems [120]. Yang et al. [126] comprehensively reviewed battery energy storage system (BESS) sizing approaches, including probabilistic methods, analytical methods, directed search-based methods (including mathematical optimization-based methods and heuristic methods) and hybrid methods. However, the BESS sizing is multi-objective optimization and decision-making with multiple systems simulation. There is no one approach outstanding others, and different approaches show advantages in different perspectives [127].

Table 4 comprehensively summarizes the distributed electrical energy systems with capacity sizing approaches, mainly including parametrical analysis, Monte Carlo simulation, single-objective and multi-objective optimizations. Regarding renewables–buildings–vehicles energy systems, Zhou et al. [128] conducted parametrical analysis on battery capacity and charging/discharging power to achieve the optimal equivalent CO_2 emissions, import cost, energy flexibility, and battery relative capacity. To address the conflictions on these objectives, multi-objective optimization was conducted with Pareto optimal solutions [21]. Furthermore, considering parameters uncertainty, Chebabhi et al. [129] conducted the Monte Carlo simulation to design and size renewable microgrid. Niveditha and Singaravel [130] optimized the hybrid PV–Wind–Battery storage system to enhance the grid independence and reduce cost. By using the Non-dominated Sorting Genetic Algorithm II (NSGA-II) to optimize annualized cost of the system (ACS), Loss of Power Supply Probability (LPSP) and total energy transfer (TET), the optimal Pareto front can be identified with optimal sizing on the integrated renewable systems.

4.4. AI-assisted energy storage approach

In addition to the traditional method for battery capacity sizing, AI-assisted energy storage sizing approach gradually emerges. Generally, the AI-assisted energy storage sizing approach mainly includes surrogate model development, performance prediction, and optimization. Dan et al. [135] developed surrogate models to replace traditional physical models of energy consumption, renewable energy supply, and battery cycling aging. The process can speed up the simulation process with accuracy in multi-scale and cross-scale modelling of 'source-grid-

Table 2
Summary of optimization and assessment criteria of renewable-storage systems.

Studies	Systems	Methods	Assessment criteria	Results
Amini et al. [112]	Battery energy storage in a microgrid	Mixed-integer linear programming	Service life, and capacity degradation	Battery energy storage (BES) capacity degradation affects BES performance and microgrid total costs.
Mair et al. [113]	Domestic batteries in buildings	Comparative analysis	Peak shaving and economic performance	Coordinated battery deployment can improve economic performances.
Novoa et al. [114]	Renewable-battery system	Mixed Integer Linear Program (MILP) optimization	Renewable penetration	High renewable penetration through the sized capacity.
Atia and Yamada [115]	Renewable-battery with electric vehicles	Stochastic modelling	Economic effect	RES sizing is dependent on V2G operation.
Alberizzi et al. [116]	Hybrid Renewable Energy System with battery storage.	Mixed Integer Linear Programming (MILP) optimization	Costs and energy deficits	Cost reduction of 57 % with a maximum cost difference of 134,787 €
Kahwash et al. [117]	Grid-connected hybrid renewable energy systems	Genetic algorithm-based multi-objective optimisation	Levelised costs of electricity and heat	Levelized cost of electricity and heat can be reduced through the optimisation, while gas and electricity prices significantly affect optimum solution.
Wang et al. [118]	Energy storage with renewables	Multi-stage optimization	Renewable generation utilization	Storage investment can improve renewable self-consumption.

demand-storage' systems. Furthermore, Zhou [19] developed an AI-based battery aging prediction model with efficient and accurate predictions. Roles of machine learning e in energy storages mainly include battery capacity sizing, optimal scheduling and energy management, dynamic performance predictions (like battery state-of-charge, lifetime estimation, fault detection and diagnosis) [136].

4.4.1. Big data training surrogate model

In terms of big data training for surrogate models, Wei et al. [137] developed a data-driven optimal sizing approach to size the photovoltaics-electric vehicle charging microgrid. The role of ML is to establish the correlation between load features and economic parameters. Economic analysis results can assist the technical investment behaviours, i.e., energy storage system price lower than 77 $/kWh. Bilayered Neural Network model is developed to predict the levelized cost of electricity, so as to identify the remote location of hybrid renewable systems [138].

Fig. 6 demonstrates the surrogate model development with forward propagation and backpropagation. Energy storage sizing and analysis are shown in Fig. 6(c). In order to identify the optimal battery capacity, inputs include renewable types, renewable capacity, power grid and energy demands. Constraints include time resolution and energy prices. Depending on different optimal objectives, the battery capacity can be sized.

4.4.2. Sizing optimization with AI

AI-assisted energy storage sizing optimization will become a hot topic. Lin et al. [140] developed a deep learning-based battery sizing optimization tool to optimize the battery capacity. The proposed approach can achieve the optimal capital and operations costs, assisting the investment behaviours in battery. Kang et al. [141] studied capacity sizing and real-time scheduling of battery energy storage based on reinforcement learning-based multi-objective optimization. The objectives include total cost peak and demand decrease. Ghandehariun et al. [142] optimized renewable-energy-based multigeneration system with machine learning. The optimization approach can achieve the minimum levelized cost of electricity at $0.83/kWh.

The role of the machine learning is for accurate performance predictions of nonlinear energy systems.

Machine learning plays significant roles in surrogate model development, so as to be integrated with optimization algorithm for optimal system design energy [143]. Khan and Javaid [144] optimized PV-wind turbine with battery systems based on Jaya Learning. A cost-effective solution can be provided based on the proposed optimization approach [145]. Toosi et al. [146] developed a machine learning-based optimization approach for energy storage design.

Battery sizing optimization with AI is shown in Fig. 7. The underlying mechanism of AI-assisted energy storage approach for capacity sizing can be summarized below. Techno-economic performances can be predicted through the AI-based surrogate model with inputs of battery capacity. Afterwards, the AI-based surrogate model is integrated in the optimization engine to search for the optimal value of battery capacity, which can maximize the techno-economic performances. Specifically, the collected data can be used to train, test and validate the surrogate model, as shown in Fig. 7. Afterwards, the optimization engine is to call the surrogate model time by time to timely get the performance results. During the optimization process, various optimization algorithms can be used and the optimal results will be algorithm-dependent. Future studies can focus on advanced algorithms development for fast and accurate performance predictions. Moreover, surrogate models with human knowledge-based machine learning need to be developed.

4.4.3. Research limitations

Based on above literature review, research limitations are summarized below:

1) the current studies fail to consider battery aging performance prediction during the battery capacity sizing process. The only consideration on operating costs without initial investment costs and depreciation cost will lead to overestimation.
2) by developing the surrogate model for battery performance predictions, it is really difficult to predict the performance when parameters are out of the range of the training database. Uncertainty analysis and optimization may be required for robust optimal results on battery storages.
3) due to the diverse techno-economic-environmental performance criteria, there is no agreement on which criterion needs to be focused. The lack of standard or method leads to the diversified optimal design results.

5. Techno-economic analysis (TEA) for renewable-storage-grid energy systems with optimal sizing

Renewable-storage-grid energy systems have been studied with optimal sizing for techno-economic performance improvement. Table 5 summarizes techno-economic performance optimization for renewable-supported energy systems. As listed in Table 5, to help achieve the optimal objectives (e.g., net present cost, cost of energy, the loss of power supply, Life Cycle Cost, etc.), various optimization approaches can be applied.

Ashtiani et al. [148] optimized the grid-connected PV/battery system with the teaching-learning-based optimization algorithm. Through

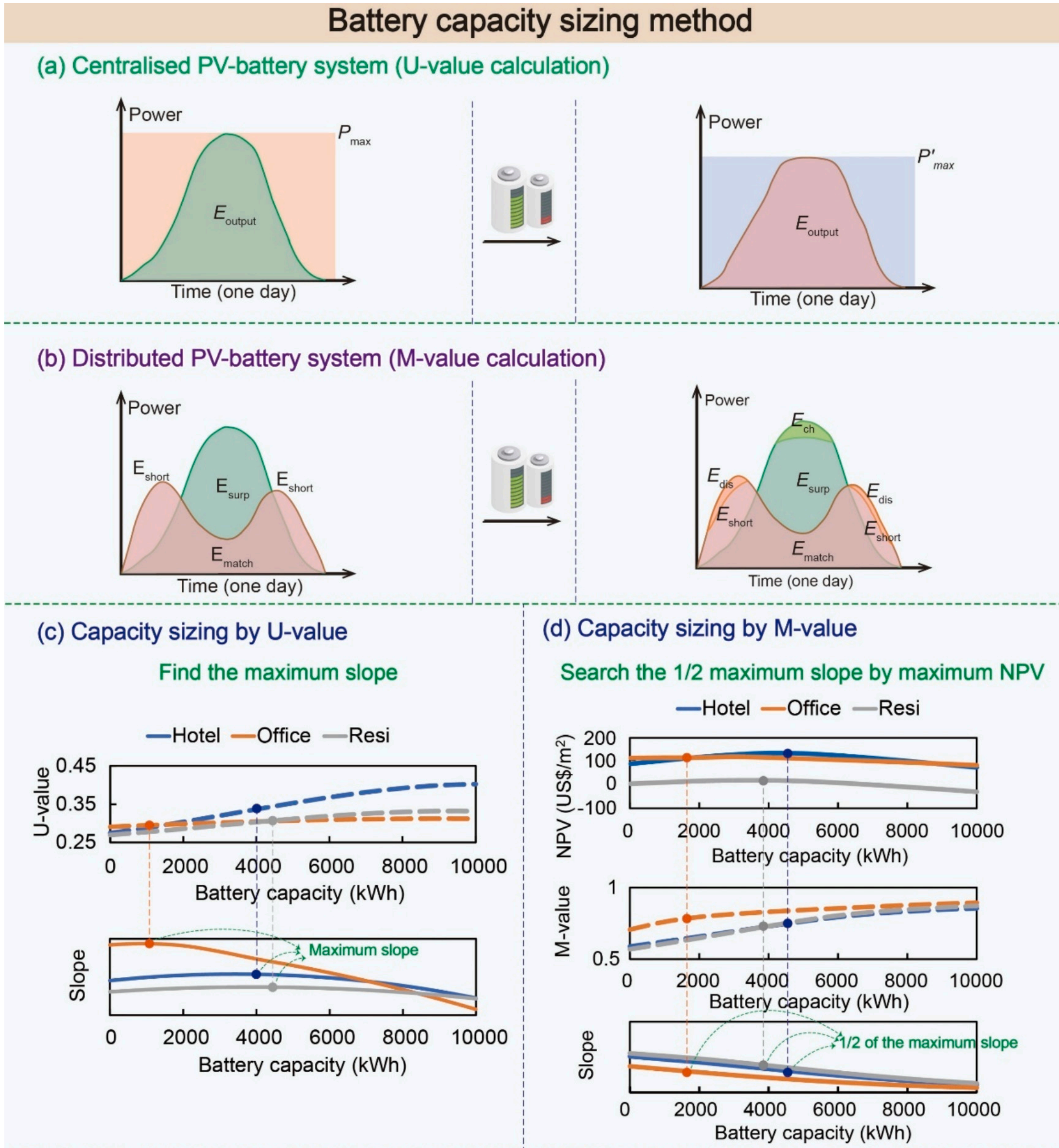

Fig. 5. The battery capacity sizing methods. (a) U-value calculation for centralized PV-battery-consumer system; (b) M-value calculation for distributed PV-battery-building system; (c) battery capacity by U-value approach in the centralized PV-battery-consumer system; (d) battery capacity by M-value approach in the distributed PV-battery-building system [120]. Figure 5 is partially reprinted and partially redrawn from Song and Zhou [120] permission from Spring Nature.

Table 3
Centralized electrical energy systems with capacity sizing.

Studies	Systems	Sizing approaches	Assessment criteria	Results
Al-Masri et al. [121]	Hybrid PV/wind//pumped hydro storage systems	Flow direction optimization algorithm	Renewable storage factor and levelized cost of energy	The best design is identified as hybrid PV/wind/pumped hydro storage (PHS) systems.
Kahwash et al. [117]	Grid-connected hybrid renewable energy systems	Genetic algorithm-based multi-objective optimisation	Levelized cost of electricity and heat	Levelized cost of electricity and heat can be reduced through the optimisation, while gas and electricity prices significantly affect optimum solution.
Borkowski et al. [123]	Photovoltaic farm with battery storage	Energy management and optimization	Daily market revenue	Increase in the rate of return of the energy storage with increase in renewable penetration.
Sadeghibakhtiar et al. [122]	Stand-alone solar-wind-battery hybrid system	Genetic algorithm-based single-objective approach and NSGA-II based double-objective approach	Energy supply reliability and system costs.	Compared to single-objective optimisation results, the double-objective optimization achieves a lower cost with a lower power supply reliability.
Ibáñez-Rioja et al. [124]	Solar PV–wind power–battery–water electrolyzer plant	Parametrical analysis	Levelized cost of hydrogen	Wind based system is the most cost-effective solution and the levelized cost of hydrogen is less than 2 €/kg by the 2030.
Altin [125]	Photovoltaic-Wind-Battery-Inverter system	Parametrical analysis	Cost of Energy	System sizing is dependent on renewable energy sources.

Table 4
Distributed electrical energy systems with capacity sizing.

Studies	Systems	Sizing approaches	Assessment criteria	Results
Sun et al. [131]	Virtual power plants	Non-cooperative generalized Nash game	Frequency regulation service provision	Efficiency loss can be effectively reduced from around 80 % by increasing demand-side resource population from 20 to 2000.
Al-Quraan and Al-Mhairat [132]	Standalone hybrid renewable energy systems	Bi-level mixed integer nonlinear programming	Techno-economic-based multi-objective function	The lowest total cost and energy loss facto at around 132,000 $ and 0 with energy loss factor less than 0.0009.
Chebabhi et al. [129]	Renewable energies with photovoltaics/wind/diesel in microgrids	Monte Carlo simulation	Net Present Cost, Levelized Cost of Energy and Discounted Payback	Financial analysis plays critical roles in the design and sizing of RES microgrid.
Elfatah et al. [133]	Hybrid PV/diesel/battery systems	Flow direction algorithm	Cost of energy, net present cost	The proposed modified Flow Direction Algorithm show superiorities in designing hybrid microgrid component sizing.
Zhou and Xu [134]	Stand-alone microgrid system with Photovoltaic arrays (PV), Wind turbines (WT), Tidal turbines (Tid), Battery (Bat) storage and hydrogen storage.	Chameleon swarm algorithm optimisation	Net present cost and Loss of power supply probability	PV/WT/Tidal turbines/Battery system is the most optimal approach for local load supply.
Zhou et al. [128]	Renewables–buildings–vehicles energy systems	Parametrical analysis	Equivalent CO_2 emissions, import cost, energy flexibility, and equivalent relative capacity of battery storage.	Multi-criteria can be improved through the proposed strategy.
Zhou et al. [21]	Buildings-vehicles energy system	Multi-objective optimisation with Pareto archive NSGA-II algorithm	Cost, emission and energy flexibility	Optimal solutions can be provided to address energy-related conflicts.

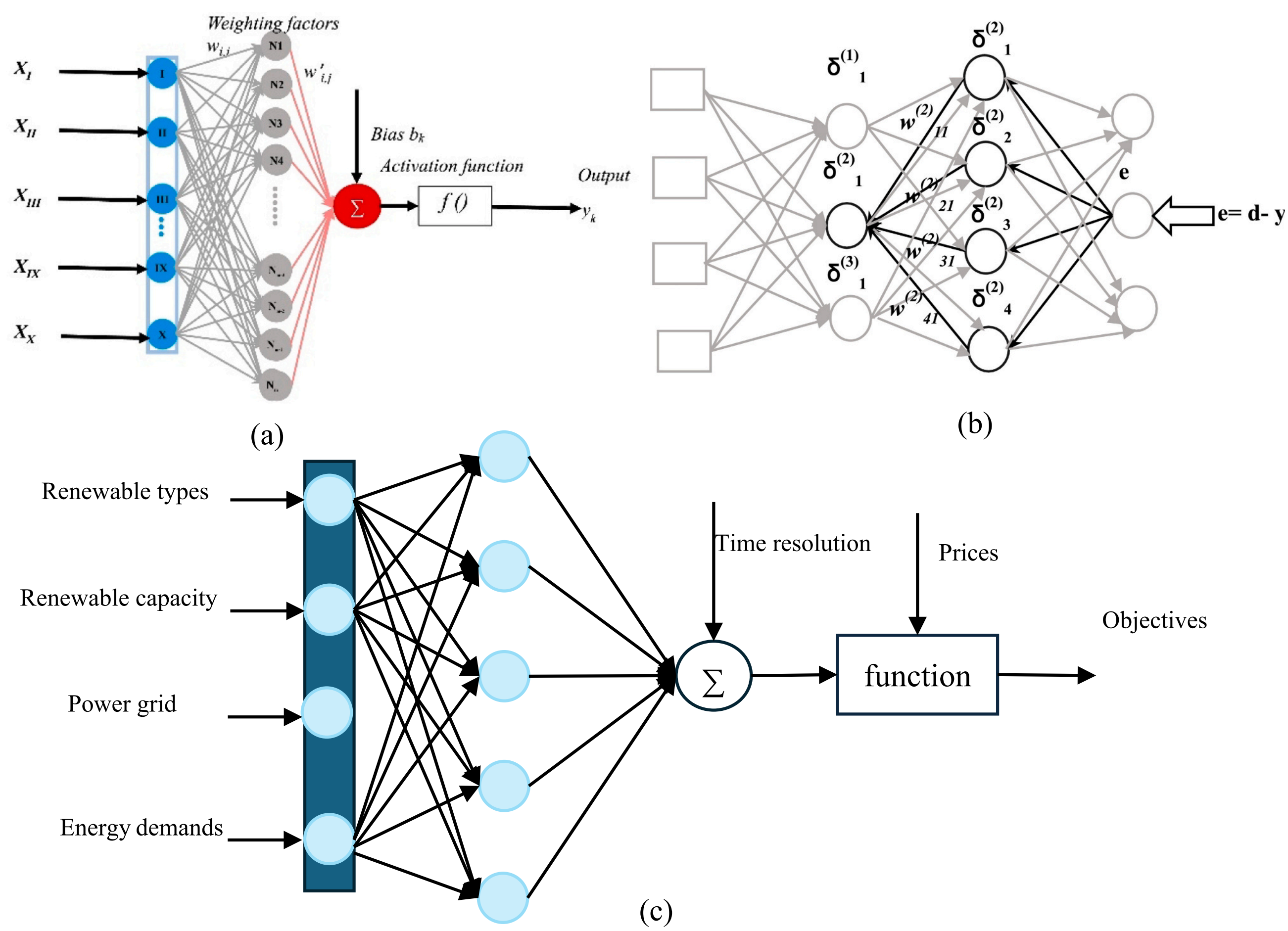

Fig. 6. Surrogate model development on energy storages: (a) forward propagation; (b) backpropagation [139]; (c) energy storage sizing and analysis. Figure 6 is partially reprinted and partially redrawn from Zhou [139] with permission from Elsevier.

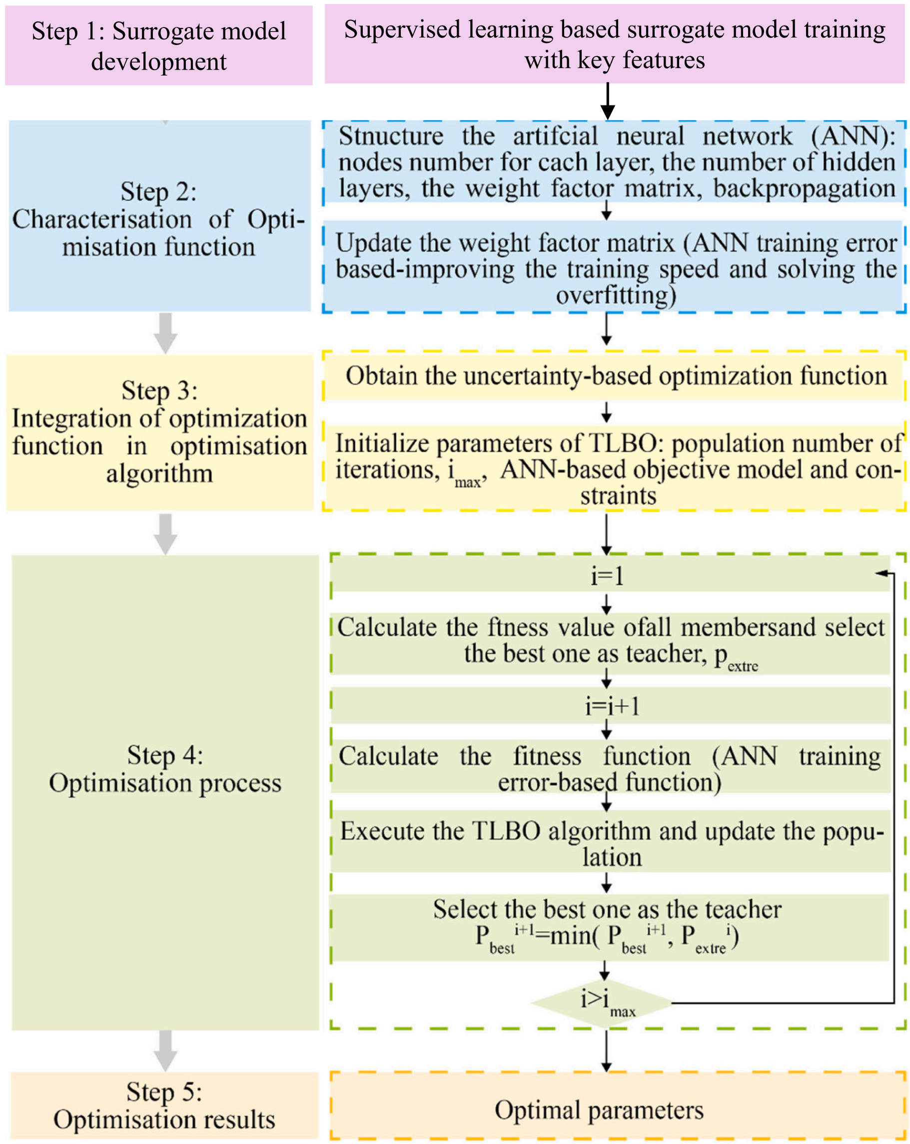

Fig. 7. Battery sizing optimization with AI [147]. Figure 7 is partially reprinted and partially redrawn from Zhou and Zheng [147] with permission from Elsevier.

the proposed optimisation approach, cost of energy and net present cost can be improved by 8.1 % and 6.7 %, respectively. Furthermore, considering conflictions between the loss of power supply and the demand coverage from renewables, Parvin et al. [149] conducted multi-objective optimization based on particle swarm algorithm. Results indicated that the optimum loss of power supply was 0.207 with the improvement of the demand load coverage by 20 %. Mohammed et al. [150] conducted tri-objective techno-economic optimization on Life Cycle Cost, Levelized Cost of Electricity and Loss of Power Supply Probability, so as to achieve the optimal operating and autonomy of the system.

6. Comparison with other studies, outlook and recommendations

Battery cycling aging for E-mobility-based circular economy has been quantified with battery replacement times over 25 years [153]. Use-phase carbon footprint of EVs has been studied [154]. Carbon footprint of energy storages [155] and lifecycle carbon footprints of battery have been studied [156,157], while the optimal battery capacity has not been provided, leading to the battery capacity oversizing and performance overestimation. This study highlights the necessity of battery capacity sizing for carbon footprints and techno-economic performance. As an intermediate bridge between energy supply and energy demands, energy storages are essential for renewable energy penetration, load coverage, power supply reliability, grid power stabilization, and so on. Future studies can focus on:

1) In addition to battery capacity sizing, optimal battery energy management plays significant roles in operating cost saving and battery service lifetime extension. Machine learning-based predictive energy management strategy with multi-objective optimizations needs to be studied [158];
2) In terms of the integrative energy system, carbon intensity database with both embodied and operating carbon emissions needs to be established, to guide the carbon analysis and comparison;
3) Climate adaptive battery sizing approaches need to be developed and frontier guidelines for battery capacity design are necessary with techno-economic-environmental performance analysis, like battery carbon intensity [156].

7. Conclusions

This study focuses on renewable-storage sizing approaches for

Table 5
Summary of techno-economic optimization with renewable-supported energy systems.

Studies	Systems	Optimization approaches	Objectives	Results
Ashtiani et al. [148]	Grid-connected PV/battery system	Teaching-learning-based optimization algorithm	Net present cost (NPC) and cost of energy (COE)	Through the proposed optimisation approach, cost of energy and net present cost can be improved by 8.1 % and 6.7 %.
Parvin et al. [149]	Renewable micro grid	Multi-objective particle swarm optimization algorithm	The loss of power supply and the share of demand load supplied by renewable	The optimum loss of power supply is 0.207 and the share of demand load supplied by renewable can be increased by 20 %.
Merei et al. [151]	A PV-battery system	Genetic algorithm	Self-consumption, self-sufficiency and electricity costs	Batteries are not economic in improving self-consumption.
Shabani et al. [152]	Grid-connected PV system	Non-dominated sorting genetic algorithm	Battery life cycle cost and the self-sufficiency ratio	Accurate battery sizing is significant in determining the techno-economic performance.
Mohammed et al. [150]	Solar Photovoltaic/ Battery system	Tri-objective techno-economic optimization	Life Cycle Cost, the Levelized Cost of Electricity and the Loss of Power Supply Probability	Optimal operating and autonomy of the system can be ensured through the proposed approach.

centralized and distributed renewable energy systems to avoid the battery capacity oversizing or under-sizing and resource wastes. Roles of centralized and distributed energy systems are characterized in low-carbon transitions. Renewable-storage sizing of both centralized and distributed renewable-storage systems are characterized by ‘U-value’ approach and ‘M-value’ approach, respectively. Lastly, AI-assisted energy storage approach is also prospected with big data training surrogate model and sizing optimization with AI. Techno-economic analysis (TEA) for renewable-storage-grid energy systems under the optimal sizing has been conducted. Key conclusions are summarized below:

1) Unlike centralized PV-battery-consumer systems that mainly focus on intermittent renewable energy, energy storages in distributed prosumer-battery systems have to dynamically balance on-site renewable energy supply and energy demand [119], imposing challenges battery capacity optimization. However, in terms of electrified lifecycle sustainable transformation, whether a centralized or distributed energy system is the most optimal design solution is still questionable.
2) Compared to centralized energy systems, distributed energy systems are more flexible in power sharing, transmission and distribution. Furthermore, distributed energy systems can enable self-consumptions to reduce the energy storage capacity and enable fast demand response and recovery with high energy resilience when suffering from nature disasters. By contrast, centralized energy systems show a higher energy efficiency, power supply reliability, and etc.
3) The proposed ‘U-value’ and ‘M-value’ approaches are effective in sizing battery capacity for centralized and distributed PV-battery-consumer systems, respectively. The general principle of the former is to stabilize the battery capacity-dependent outpower from the battery. The principle of the latter is to reduce the mismatched energy (i.e., surplus renewable energy, and demand shortage).
4) Techno-economic and life cycle assessment on energy storage technologies is critical for capacity sizing. Multiple assessment criteria mainly include renewable penetration, battery capacity degradation and service life, levelized costs of electricity and heat, and so on. In addition to the traditional method for battery capacity sizing, AI-assisted energy storage sizing approach gradually emerges. Generally, the AI-assisted energy storage sizing approach mainly includes surrogate model development, performance prediction, and optimization.

Future studies can focus on optimal battery energy management for operating cost saving and battery service lifetime extension, machine learning-based predictive energy management strategy and national/ global carbon intensity database establishment. Furthermore, climate-adaptive battery sizing approaches need to be developed with frontier guidelines for battery capacity design.

CRediT authorship contribution statement

Yuekuan Zhou: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declared that they have no conflicts of interest to this work.

I declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

The data that has been used is confidential.

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